

Name: _____



Homework Questions for Physics and Reality Week 3

Due the week of October 13

SHOW ALL WORK

Metric System Prefixes Table

T	“terra”	10^{12}
G	“giga”	10^9
M	“mega”	10^6
k	“kilo”	10^3
c	“centi”	10^{-2}
m	“milli”	10^{-3}
μ	“micro”	10^{-6}
n	“nano”	10^{-9}
p	“pico”	10^{-12}

Useful Values:

Planck’s constant: $h = 6.63 \times 10^{-34} \text{ J s}$

Mass of an electron: $m_e = 9.11 \times 10^{-31} \text{ kg}$

Speed of light in a vacuum: $c = 3.00 \times 10^8 \text{ m/s}$

$1 \text{ J} = 1 \text{ kg m}^2/\text{s}^2$

$1 \text{ mile} = 1.61 \text{ km}$

Useful Equations

De Broglie wavelength

$$\lambda = \frac{h}{mv}$$

λ = wavelength, h = Planck’s constant, m = mass, v = speed

Heisenberg Uncertainty Principle

$$\Delta x \Delta v \geq \frac{h}{4\pi m}$$

Δx = quantum uncertainty in position, Δv = quantum uncertainty in speed, h = Planck’s constant, m = mass

1. Double Slit Experiment, Wave Particle Duality, and the Uncertainty Principle

As Professor Greene described in his lecture this week, the double slit experiment teaches us that microscopic objects, like electrons, are both particles and waves. In fact, Louis de Broglie suggested in 1923 that all material objects display wave-like properties and have wavelengths associated with them.

According to de Broglie, these “matter waves” have a wavelength given by the equation:

$$\lambda = \frac{h}{mv}$$

where λ is the wavelength of the object, h is Planck’s constant, m is the mass of the object, and v is its speed.

Let’s consider what this formula and its application tell us about the electrons in the double slit experiment.

As a reminder, the experiment is set up such that you send various objects through an impenetrable wall containing two slits and those objects hit a detecting screen, as shown in Figure 1 below.

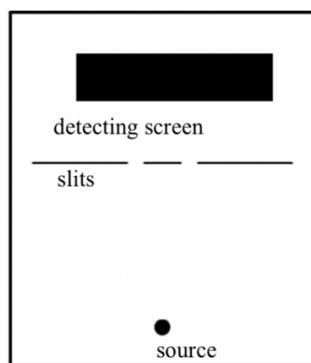


Figure 1. *Sketch of the setup for the double slit experiment.*

1A. (i) Suppose that in your double slit experiment, your electrons are moving at a speed of 850 m/s. **What is the wavelength of these electrons in units of nm?**

1A. (ii) Do the electrons in the double slit experiment exhibit any particle-like behavior, and if so, when? If YES circle the location(s) where the electrons behave as particles in Figure 1 (previous page). If NO explain in 1-2 sentences.

1B. When an object's wavelength is small relative to its environment, it behaves like a particle. When an object's wavelength is comparable to or large relative to its environment, it behaves like a wave. In 2012 researchers at the University of Nebraska-Lincoln performed the double-slit experiment, exactly as it was described in the lecture. Electrons were fired by an electron gun and passed through a wall with two slits. The slits were 272 nm apart from each other.

1B. (i) Compare the wavelength that you calculated in question 1A. (i) with the slit separation used in the University of Nebraska-Lincoln experiment. **Based on this comparison, what kind of behavior do you think the electrons will manifest as they approach the slits?** Explain your choice in one sentence.

- a. Wave-like
- b. Particle-like
- c. Neither wave-like, nor particle-like

1B. (ii) Given your answer to part 1B. (i) above, which slit does each electron pass through? Explain your choice in one sentence.

- a. One of the slits only, which we could determine if we had a sufficiently precise measuring apparatus
- b. One slit only, but we can never determine which one
- c. Both slits at the same time

In the lecture, you also learned about the Heisenberg Uncertainty Principle ($\Delta x \Delta v \geq \frac{h}{4\pi m}$).

According to this principle, there is a limit on the product between the uncertainty in position (a particle-like quality of the object) and the uncertainty in speed (a wave-like quality of the object, since, as de Broglie showed, speed and wavelength of the object are related). Let's consider what we can learn from the Heisenberg Uncertainty Principle both qualitatively and quantitatively.

1C. Suppose that the scientists conducting the experiment at the University of Nebraska-Lincoln, did a version of the experiment in which the electrons were sent out towards the slits one by one. Furthermore, suppose that they wanted to determine exactly which slit each electron went through. In order to do so, one would need to determine the position of the electron to within the width of each slit. In other words, the uncertainty in the position of the electrons would equal the width of the slits. In their setup, the width of each slit was 65 nm.

1C. (i) What would be the minimum uncertainty in the speed of the electrons as they passed through the slit, given an uncertainty in position of 65 nm?

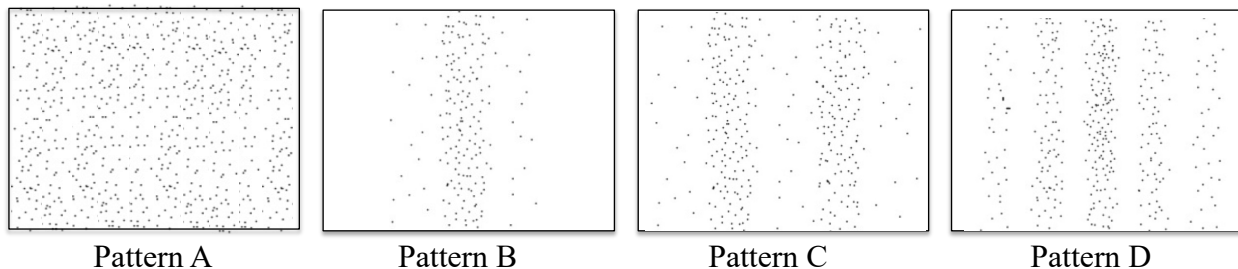
1C. (ii) Is the uncertainty in the electron's speed small or large? **To answer this, compare the uncertainty you calculated in part 1C. (i) above to the electron's speed (850 m/s).** Report your answer as a percentage.

1C. (iii) Recall that a particle has a very well-defined position, whereas a wave has a very well-defined speed. **When experimenters determine which slit the electrons pass through, do the electrons behave in a wave-like or particle-like fashion?** Select the best answer.

- a. particle-like, because, once experimenters determine which slit electrons pass through, the uncertainty in the position of the electrons is zero.
- b. particle-like, because the uncertainty in the speed of electrons is large compared to their speed.
- c. wave-like, because the uncertainty in the speed of electrons is large compared to their speed.
- d. wave-like, because electrons in the double-slit experiment always behave wave-like before they hit the screen.
- e. neither wave-like, nor particle-like, because we can never know anything about both the position and speed of the electrons.

1C. (iv) Based on your answer to question 1C. (iii), what kind of pattern would researchers observe on the screen when they try to determine which slit the electrons pass through?

Circle the best pattern and explain your choice in one sentence.



1D. Suppose that instead of electrons we use marbles in the same setup as in Figure 1. Note that the size of the slits would have to change for the marbles to fit through, but we would still like to determine the position of the marbles to within 65 nm. The mass of a marble is ~ 15 g.

1D. (i) What would be the minimum uncertainty in the speed of the marbles?

1D. (ii) Is the uncertainty in the marble's speed small or large? To answer this, compare the minimum uncertainty in the speed of an electron from part 1C. (i) to the minimum uncertainty in the speed of a marble from part 1D. (i). **Which is larger and by how many orders of magnitude?**

1E. Consider your answers to questions 1C and 1D above. **What do they suggest about quantum mechanics more generally, as it applies to microscopic vs. macroscopic objects?** Select the best answer.

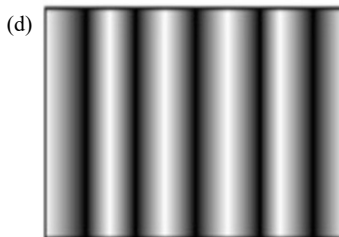
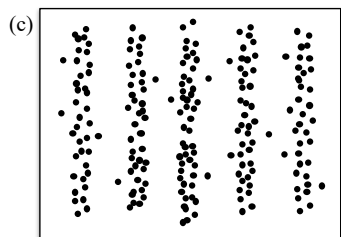
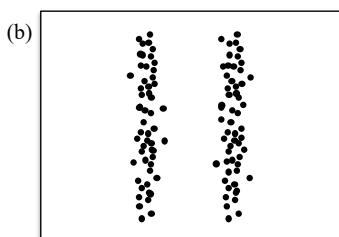
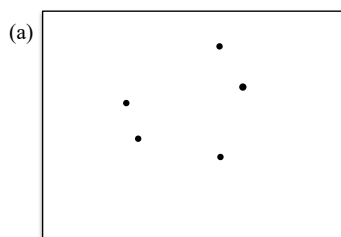
- Quantum mechanics only applies to microscopic objects, because macroscopic objects do not exhibit a wave-like nature.
- Quantum mechanics applies to both microscopic and macroscopic objects, but its effects are extremely small and thus negligible for macroscopic objects.
- Quantum mechanics only applies to elementary particles, such as electrons.
- Quantum mechanics does not really apply to any type of particle, because, as we learned in lecture, it does not agree with General Relativity.

1F. In the previous questions, you determined quantitatively why objects with small vs. large masses behave the way they do in the double-slit experiment. Now let us examine qualitatively the kinds of behaviors we would expect.

Below, there is a list of different versions of setting up the double slit experiment. Figure 2 below shows four possible screen intensity patterns. **For each intensity pattern (a) through (d), name all of the possible versions from the list that could have produced that pattern.** E.g., pattern (d) can only be produced by water waves (version 1), so in the list next to the figure you enter: *Pattern (d): 1*

Different versions of the double slit experiment:

1. Water waves pass from the source through the double slit.
2. Electrons pass from the source through the double slit.
3. Electrons are fired from the source; a detector is placed behind one of the slits, which lights up every time an electron goes through that slit.
4. Electrons are fired from the source; one of the slits (chosen at random) is blocked as each electron is fired, such that each slit is only open half of the time.
5. Marbles pass from the source through the double slit.
6. Marbles are fired from the source; one of the slits (chosen at random) is blocked as each marble is fired, such that each slit is only open half of the time.



Pattern (a): _____

Pattern (b): _____

Pattern (c): _____

Pattern (d): _____

Figure 2. Four possible patterns that might be detected on the screen.

1G. Suppose that you utilize again the basic setup of the double slit experiment described in the introduction to Question 1 and shown in Figure 1. You send millions of electrons through the wall with two slits. **From the choices given on the next page, select ALL the statements that correctly describe what we can and/or cannot predict.**

- a. We can definitively predict where an individual electron will land on the screen.
- b. We can definitely predict where an individual electron will NOT land on the screen.
- c. We can definitively predict where an individual electron will land on the screen only if we can determine precisely both the position and the speed of the electron right before it enters the slits.
- d. We can predict where an individual electron is most likely to land, but not where it will definitively land.
- e. We cannot make any predictions whatsoever about individual electrons.

2. Entanglement

In the lecture, Professor Greene discussed entanglement, using the example of two particles whose spins are entangled. The particles are in a superposition of “particle 1 spin UP, particle 2 spin DOWN” and “particle 1 spin DOWN, particle 2 spin UP.” He also described that even after the entangled particles are separated by very large distances, measuring the spin of one particle immediately affects the spin of its entangled partner. For example, if one measures particle 1 and finds it is spin DOWN, particle 2, no matter how far, will immediately assume the spin UP state.

2A. Do you think that this "spooky action at a distance" violates the principle that nothing travels faster than the speed of light? State Yes or No and explain in one to two sentences.

2B. Professor Greene decides to demonstrate entanglement to Frontiers of Science students during lecture. First, he uses a pair of gloves. He takes two identical boxes and puts one glove in each box. He randomly gives one box to Ivana sitting in the front and passes one box to Daphna sitting in the back of the lecture hall.

2B. (i) Suppose that Ivana got the box first and is about to open it, while Daphna in the back is still waiting for her own box to reach her. **Which of the following is true when Ivana is about to open her box? Pick ALL the correct statements.**

- a. The glove in Ivana's box is either the left glove or the right glove, but not both.
- b. The glove in the Daphna's box is either the left glove or the right glove, but not both.
- c. The glove in Ivana's box is both the left glove and the right glove at the same time.
- d. The glove in Daphna's box is both the left glove and the right glove at the same time.
- e. If Ivana is told that her box contains only one glove, she cannot definitively predict whether she will get the right glove or the left glove.
- f. If Ivana is told that her box contains only one glove, she can definitively predict whether she will get the right glove or the left glove.

2B. (ii) Suppose that Ivana found the left glove inside her box but has not told anyone yet. **Which of the following is true when Daphna is about to open her own box?** Pick ALL the correct statements.

- a. The glove in Daphna's box is definitely the right glove.
- b. The glove in Daphna's box is both the left glove and the right glove at the same time.
- c. If Daphna is told that her box contains only one glove, she can definitively predict whether she will get the right glove or the left glove.
- d. Daphna can determine whether she has the left glove or the right glove, only if she is told what Ivana found in her box.

2C. Professor Greene next entangles two electrons, such that they are in a quantum superposition of two spin states: "electron 1 UP, electron 2 DOWN" and "electron 1 DOWN, electron 2 UP," each state with 50% probability. He takes two boxes and places one entangled electron in each box, followed by giving one box to a student on his right and one box to a student on his left. Professor Greene then asks the students to open their boxes and record the electron spin, while he is not looking. Finally, he asks the student to his right to show him what they wrote down. Their note reads: "Electron spin: UP."

2C. (i) Knowing just this piece of information, can Professor Greene determine what the student to his left wrote down?

- a. NO. Professor Greene did not know the order in which the students to his left and his right opened their respective boxes.
- b. NO. Professor Greene (or anyone) cannot make definite predictions for microscopic particles such as electrons.
- c. YES. The student wrote "Electron spin: DOWN", because that electron was always spin DOWN.
- d. YES. The student wrote "Electron spin: DOWN", because the other electron was found to be spin UP.